

Blockchain-based Digital Money for Bank Indonesia Business Process

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Abstract—Central Bank Digital Currency (CBDC) is a strategic Bank Indonesia initiative under Project Garuda, but the publicly documented Garuda experiments focus on the wholesale tier, leaving a technical gap for a testable system that digitizes Bank Indonesia’s retail money business processes—issuance, two-tier distribution, risk-based customer onboarding, payment, redemption, and supervision—under one auditable blockchain ledger. This paper designs and implements such a system: a retail Digital Rupiah prototype on a permissioned five-organization Hyperledger Fabric network. Go chaincode stores only KYC status, risk level, due-diligence level, senior-approval status, expiry, and document hashes, while raw identity data remains in an off-chain PostgreSQL store. Retail wallet tiers (BASIC, STANDARD, MERCHANT) are derived deterministically from the approved KYC profile rather than selected by the user, and each tier enforces per-transaction, daily-outgoing, monthly-outgoing, monthly-incoming, and maximum-balance limits computed from the Fabric transaction timestamp for endorsement determinism. Institutional Bank Indonesia, bank, and payment-service-provider wallets remain outside retail tiering and use system-level limits. We formalize the transfer-validation predicate, the tier-derivation function, and the double-spend-safety argument, and evaluate performance with a reproducible Hyperledger Caliper protocol that sweeps client concurrency and exercises the business processes directly. Across transfer rounds at three concurrency levels and four business-process workloads the prototype processed 616 transactions with zero failures, zero MVCC read conflicts, and zero policy or limit rejections. Transfer throughput saturates near 14.5 TPS independent of client concurrency while latency stays bounded (mean ≤ 0.81 s, maximum ≤ 2.32 s), and monetary operations reach 19.9 TPS at 0.32 s mean latency. The result is an auditable technical basis for studying how Bank Indonesia’s business processes can be carried on blockchain-based digital money without claiming production readiness.

Index Terms—Digital Rupiah, retail CBDC, Hyperledger Fabric, risk-based KYC, wallet limits, business process, blockchain benchmarking

I. INTRODUCTION

Central banks worldwide are evaluating Central Bank Digital Currency (CBDC) as a digital form of sovereign money [1]. In Indonesia, Bank Indonesia pursues this agenda through Project Garuda, whose publicly documented proof of concept targets the *wholesale* tier: a cash ledger for issuance and redemption among the central bank and large financial institutions [2], [3]. The retail tier—the distribution of Digital Rupiah to households and merchants through intermediary banks and

payment service providers (PJP)—is described at the level of policy and architecture in the Garuda White Paper [3], but no openly available, testable system carries Bank Indonesia’s retail money business processes end to end on a blockchain ledger.

Digital money for a central bank is not a single ledger operation but a chain of business processes: the central bank *issues* money against reserves, banks *distribute* liquidity downstream to PJP, PJP *onboard* customers and merchants under risk-based Know-Your-Customer (KYC) checks, holders *transact* within category-specific limits, intermediaries *redeem* money back to the central bank, and the supervisor *reconciles* and audits the whole circulation. A blockchain-based realization must encode each of these processes as deterministic, independently validated transactions while preserving two-tier intermediation that protects banks from disintermediation [4] and exposing an audit surface to the supervisor. Prior technical work tends to isolate single concerns: permissioned CBDC architecture [5], privacy designs [6], security threat models [7], or raw throughput on account-based systems [8]. Deployed retail systems such as Nigeria’s eNaira [9], China’s e-CNY [10], and the Bahamian Sand Dollar [11] publish design papers but not reproducible performance benchmarks, and the surveyed literature lacks an integrated, openly measurable retail reference for the Indonesian two-tier model [12], [13].

This paper addresses that gap with a Digital Rupiah prototype on a permissioned Hyperledger Fabric [14] network that maps Bank Indonesia’s business processes onto chaincode transactions. The contributions are:

- 1) An integrated two-tier Digital Rupiah chaincode that unifies hierarchical issuance and distribution, risk-based KYC with deterministic wallet-tier derivation, tiered transfer limits, and supervisor-facing reconciliation in a single artifact, completing the retail design beyond the wholesale-only Garuda proof of concept.
- 2) A formal, deterministic, double-spend-safe transfer protocol that combines a tier-limit validity predicate, Fabric multi-version concurrency control (MVCC), and a transaction-timestamp clock, so that limit counters and validation are identical across endorsing organizations.
- 3) A reproducible open retail-CBDC benchmark with Hy-